

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

Nature-Inspired Computation

Module Leader: Dr Edward Keedwell

TWO HOURS

Answer question 1 and two out of the three other questions. Question 1 is worth 40 marks. Other questions are worth 30 marks each.

Candidates are advised to spend FORTY-FIVE minutes on question 1 and THIRTY-FIVE minutes on each of the other questions.

No electronic calculators of any sort are to be used during the course of this examination.

ECM3412 (2016)

Compulsory question

1. (a) Write pseudocode for *ant colony optimisation* (ACO). In your answer, you should describe which features of the algorithm allow the algorithm to explore the search space, exploit what has been learned and eventually converge on a solution.

(8 marks)
- (b) Some ant colony optimisation (ACO) applications require the use of a *construction graph*. Describe, with the aid of a labelled diagram, the structure of a construction graph and explain why it is needed to enable ACO to be applied to certain problems.

(6 marks)
- (c) Evolutionary algorithms make use of selection operators to select members of the population to be parents of solutions in the next generation.
 - i. Describe the process of *tournament selection* in an evolutionary algorithm.

(2 marks)
 - ii. In tournament selection describe what happens to the selection pressure as the value of the tournament size t is increased and explain why this is the case.

(4 marks)
 - iii. Explain why tournament selection is often preferred to other selection methods such as *roulette wheel selection*.

(4 marks)
- (d) Craig Reynolds' *Boids* provide an efficient simulation of flocking behaviour on a computer. Name and briefly describe the three rules, known as *Reynolds' Rules*, used in this simulation.

(6 marks)
- (e) Collectives in nature naturally exhibit a behaviour known as *emergence*.
 - i. Briefly explain what the term emergence refers to in this context.

(2 marks)
 - ii. Briefly explain why this behaviour is important for swarm intelligence algorithms such as PSO.

(2 marks)
- (f) There are many different variants of neural networks that exist in addition to the most popular type, the multilayer perceptron. For each of the following

networks, explain the key differences between the network provided and a multilayer perceptron in terms of their architecture and the learning problem they would be best suited to.

i. Perceptron

(2 marks)

ii. Self Organising Map

(2 marks)

iii. Recurrent neural network

(2 marks)

(Total 40 marks)

Please Turn Over

2. Evolutionary algorithms are excellent general purpose optimisers.

- (a) Describe the main algorithm and, briefly, the experimental findings in the paper Wloch, K and Bentley, P.J. (2004) "Optimising the Performance of a Formula One Car using a Genetic Algorithm" in the *Proceedings of PPSN Parallel Problem Solving from Nature-VIII, Lecture Notes in Computer Science, Volume 3242/2004*.

(15 marks)

- (b) Evolutionary algorithms (EA) can be complex and have a number of parameters that can be modified. When answering the following questions you should assume that a standard generational genetic algorithm is being applied to a problem of significant computational complexity.

- i. Name and describe an appropriate *mutation operator* for the following encodings/representations:

A. Permutation

(2 marks)

B. K-ary

(2 marks)

C. Continuous

(2 marks)

- ii. Explain how you would identify that *early convergence* had occurred within an EA run. Also explain the changes you would make to the algorithm parameters in response to this finding to prevent this from happening.

(5 marks)

- iii. Explain the likely effect of the following parameter settings on an EA optimisation run:

A. Mutation rate is set to 0.0 (i.e. mutation is 'switched off')

(2 marks)

B. Crossover rate set to 0.0 (i.e. crossover is 'switched off')

(2 marks)

(Total 30 marks)

Please Turn Over

3. (a) Multi-Objective Evolutionary Algorithms differ in several ways from their single-objective counterparts.
- i. Explain what is meant by the term *pareto-front* for a given population in multi-objective evolutionary algorithms.
(4 marks)
 - ii. Explain what the desirable properties of the pareto-front are and why these properties lead to better decision making for the end-user.
(6 marks)
 - iii. Explain why multi-objective optimisation is preferable to repeating a single-objective optimisation run with different objective weightings (also known as the *weighted sum approach*).
(5 marks)
- (b) Particle swarm optimisation (PSO) and ant colony optimisation (ACO) are popular optimisation algorithms that are often grouped together in the category *swarm intelligence*.
- i. Explain the similarities between PSO and ACO that allow them to be grouped in the swarm intelligence category.
(5 marks)
 - ii. Explain the differences between PSO and ACO that show that they are different algorithms.
(5 marks)
 - iii. Describe the indicators you could use to determine that a population of particles (PSO) and a population of ants (ACO) had converged on a final solution for an optimisation problem.
(5 marks)
- (Total 30 marks)

Please Turn Over

4. (a) Traditionally, perceptrons learn through the use of a learning algorithm (such as the delta rule), however, evolutionary algorithms (EAs) can also be used to optimise the weights within a perceptron.

i. Describe how you would use an evolutionary algorithm to optimise the weights in a perceptron, paying particular attention to the representation, selection process, variation operators, fitness function, parameter settings and experimental design used to test your approach.

(10 marks)

ii. The EA could also be used to optimise the weights of a multi-layer perceptron (MLP). Explain how you would change the EA representation to optimise the weights of an MLP and discuss any differences you would expect to see in algorithm behaviour as a result.

(5 marks)

- (b) *Self-organising maps* (also known as *Kohonen networks*) are neural networks that are able to learn structure from data using unsupervised learning.

i. Explain what is meant by the word *unsupervised* in this context and how *unsupervised learning* differs from *supervised learning*.

(5 marks)

ii. Show, with the aid of equations, how a self-organising map learns from an input pattern presented to it.

(5 marks)

iii. Two variables involved in the learning process in a self-organising map will change over time. Explain what these variables are, how they change, and the effect this has on the learning process.

(5 marks)

(Total 30 marks)

End of Paper